Can Rumor Detection Enhance Fact Verification? Unraveling Cross-Task Synergies Between Rumor Detection and Fact Verification

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***Abstract*—Recently, rumor detection (fake news detection) has seen a surge in research interest, and fact verification (fake news checking) has simultaneously become a significant research aspect. Despite the inherent distinction between fact verification and rumor detection — the former being a three-category task and the latter a binary one — there has yet to be in-depth exploration into the synergies between these two tasks. Furthermore, given the severe scarcity and the time-consuming and costly construction nature of fact verification datasets, few-shot/zero-shot fact verification methods are particularly favored. To tackle these challenges, we conduct a series of studies around "How can rumor detection enhance few-shot fact verification, and to what extent?". Specifically, we systematically investigate the knowledge transferability between the two tasks, proposing a framework, Det2Ver, that is applicable to both rumor detection and fact verification. Through the construction of adaptive prompt templates and prompt-tuned LLMs like T5, Det2Ver structural-level synchronizes the two tasks and utilizes the external knowledge from rumor detection to reinforce fact verification task. We demonstrate the significance and effectiveness of Det2Ver. Through the few-shot/zero-shot experiments on three widely-used datasets, compared to other LLMs prompt-tuning baselines, the Det2Ver for cross-task knowledge augmentation brings a significant improvement in macro-F1 for fact verification.**

***Index Terms*—rumor detection, fact verification, prompt learning, parameter efficient fine-tuning, cross-task knowledge augmentation.**

I. Introduction

THE rapid proliferation of the internet and social media platforms has revolutionized how people access news, allowing unrestricted access anytime and anywhere. The ease of obtaining news is counterbalanced by the challenge of discerning truth, highlighting the critical importance of fake news verification (alias. fact verification) in maintaining the integrity of information. Rumor detection is critical for safeguarding information integrity. It has been propelled by the success of deep learning and is gaining significant attention [1], [2], [3].

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Manuscript received April 19, 2005; revised August 26, 2024.

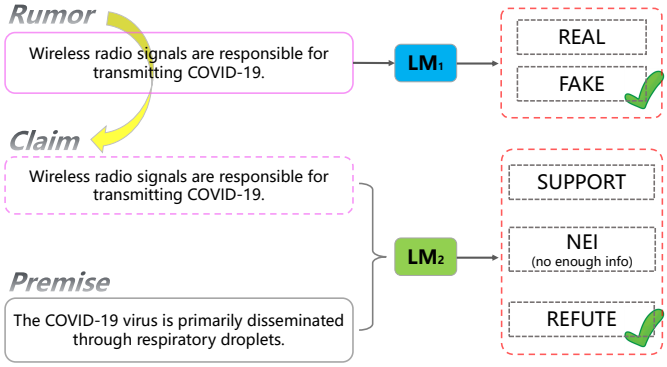


Fig. 1. The fundamental differences between Rumor Detection (Top) and Fact Verification (Bottom) tasks.

In rumor detection, the output classification is divided into a binary space $\{0, 1\}$ to determine whether the news is fake or not.

As misinformation becomes more complex, previous rumor detection methods no longer suffice, shifting focus to the increasingly vital task of fact verification, which now captures significant research interest [4].

Essentially, fact verification is a new three-category linguistic analysis task distinct from the text binary classification task of rumor detection [5]. Specifically, as depicted in Figure 1, fact verification involves a nuanced approach. Unlike directly judging the authenticity (real/fake) of news in rumor detection, it determines whether available evidence corroborates, disputes, or is insufficient to substantiate a given claim. These categories are denoted as SUPPORT, REFUTE, and NEI (Not Enough Information).

In today's information landscape, the synergy between rumor detection and fact verification is crucial. Misinformation is becoming more complex. Previous rumor detection methods are no longer sufficient. This shift has increased the focus on the vital task of fact verification, which now captures significant research interest. Fact verification involves a nuanced approach of determining whether available evidence corroborates, disputes, or is insufficient to substantiate a given claim. This shift is essential because the scarcity and high construction costs of fact verification datasets make few-shot/zero-shot fact verification methods particularly important.

Previous verification works predominantly belonged to the full-supervised learning paradigm, which have a substantial

dependence on large-scale labeled data [6], [7]. However, it is time-consuming, and labor-intensive during constructing high-quality supervised corpora, leading to a critical research bottleneck [4], [8]. Consequently, few-shot and zero-shot learning paradigms have emerged as an ever-increasing research trend in the field of fact verification. These paradigms aim to maintain competitive model generalizability under various real-life low-resource scenarios.

Until now, researchers have developed various prominent Pre-trained Language Model (PLM)-based solutions for tackling few-shot fact verification [9], [10], [11], [12]. Pan et al. [13] conducted an extensive empirical investigation on few-shot generalization in fact verification. Zeng and Gao et al. [9] proposed a novel supervised signal augmentation strategy for few/zero-shot fact verification, incorporating semantic-level consistency constraints into Pre-trained Language Models (PLMs). Zeng and Zubiaga et al. [12] developed a novel model for few-shot fact verification called Semantic Embedding Element-wise Difference (SEED). Shi et al. [14] studied cross-language and cross-domain rumor detection and proposed a superior method, which adapts from the source language and domain to the target by disentangling the semantic and emotional features of the dataset. Hou et al. [15] proposed a prompt-based approach for few-shot table-related tasks in fact verification, utilizing manually designed templates to guide pre-trained models.

Although lots of efforts were made for few- and zero-shot fact verification, they always pay attention to the optimization of syntactic/semantic-level and model architecture-level, without breaking through this bottleneck [9], [11], [12]. For instance, Could we consider the interrelationships and synergies between similar tasks, especially fact verification and rumor detection? It would be interesting also prospective to explore the synchronization and potential among these two.

Thus, in this paper, we propose a unified framework to coordinate two different tasks: Fake News Detection to(2) Fact Verification (Det2Ver). This framework aims to better utilize the sufficient supervised signal from the source task, rumor detection, to accomplish the target task of few-/zero-shot fact verification. By doing so, it gains better few-shot generalization capability.

Det2Ver achieved impressive few-shot learning performance primarily due to its unique strategies in two aspects: the construction of prompting templates and the synchronization of label mapping. Firstly, during prompt construction stage, we utilize a novel *prompting consolidation mechanism* to consolidate the original input of rumor detection and verification into a unified form. Secondly, during the label decoding stage, inspired by consistency constraints [9], we apply a novel *label words synchronization engine* to integrating outputs from both tasks into same label space.

To the best of our knowledge, it is also the first exploration to investigate the cross-task synergies between rumor detection and fact verification tasks. We are confident that this work stands as a pivotal milestone, establishing a brand-new way for future research in few-/zero-shot fact verification and potentially other linguistic tasks, thus setting a prospective direction in this domain.

Once again, the primary research motivations of this paper include the following aspects:

- Although fact verification and rumor detection have inherent differences in task nature—the former being a three-class task and the latter a binary one—the synergies between the two have not been thoroughly explored. Additionally, the scarcity and high construction costs of fact verification datasets make few-shot/zero-shot fact verification methods particularly important.
- Given the successful application of rumor detection in deep learning, exploring its knowledge transfer potential in few-shot fact verification tasks presents a promising research direction.
- Current fact verification methods mainly focus on optimization at the syntactic/semantic level and model architecture level, failing to overcome the bottleneck of data scarcity. Therefore, it is necessary to consider the interrelationships and potential synergies between tasks, especially between rumor detection and fact verification.
- By utilizing the supervisory signals from rumor detection tasks to accomplish few-shot/zero-shot fact verification tasks, we believe it is possible to enhance the generalization capability of fact verification models in low-resource scenarios.

Meanwhile, our contributions in this paper can be summarized as follows:

- We propose Det2Ver, a novel prompt-tuning framework that integrates rumor detection and fact verification tasks, leveraging the rich supervisory signals from rumor detection to enhance few-/zero-shot fact verification performance.
- Det2Ver introduces a unique prompting consolidation mechanism that unifies the input format of rumor detection and fact verification, facilitating effective cross-task knowledge transfer.
- In the Det2Ver framework, we develop a novel label words synchronization engine that aligns the output labels of both tasks into a consistent label space, enhancing the integration and performance of the unified model.
- Extensive few-/zero-shot experiments validate the superiority and robustness of Det2Ver under data-scarce scenarios, demonstrating significant improvements in few-/zero-shot fact verification.

The remainder of this paper is organized as follows. In Section II, we provide a comprehensive review of related work, highlighting the efforts and methodologies in both rumor detection and fact verification. Section III details the proposed Det2Ver framework, discussing its prompting consolidation mechanism and label synchronization strategies. Section IV describes the experimental setup and the datasets used for evaluating our framework. In Section V, we present the experimental results and analyze the performance of Det2Ver compared to existing methods. Section VI offers a discussion on the limitations and ethical considerations of our approach. Finally, Section VII concludes the paper and suggests potential directions for future research.

II. Related Works

In recent years, research related to zero- and few-shot fact verification has been booming and thriving [6]. Below are some representative works.

A. Methods

Representative methods research in this research field is currently active. Wang et al. [16] address the challenge of rumor detection on social media by proposing a novel method to assess the consistency of multimodal content using transformers and self-attention-based Bi-GRU networks. Their approach leverages the semantic comparison between images and texts within the same post to enhance rumor detection. Zeng and Gao et al. [9] devised a novel supervised signal augmentation strategy for few/zero-shot fact verification, which infused the semantic-level consistency constraints in claim-evidence pairs into Pre-trained Language Model (PLMs). Li et al. [17] explore the intersection of fact verification and fake news detection by leveraging advancements in pre-trained language models and large-scale datasets. They propose a novel method that combines text summarisation and fact verification to detect fake news effectively. Zeng and Zubiaga et al. [12] developed a novel model for few-shot fact verification, Semantic Embedding Element-wise Difference (SEED). During inference, it uses semantic difference vectors generated by PLMs to construct representative vectors that identify instance labels. Inspired by the Adversarial Contrastive Learning [18], Lin et al. [10] designed an adversarial contrastive augmentation to optimize the rumor' higher-level representation among different domains and languages. This method employed adversarial attacks for structural feature adaption to expand the low-resource data diversity, which achieves superior results for fake news classification and detection. Oni and van Zyl [19] present a comparative study of ensemble approaches to fact-checking using the FEVER shared task. This task involves Document Retrieval, Sentence Selection, and Claim Verification. They standardize TF-IDF for Document Retrieval and create pipelines for Sentence Selection and Claim Verification, evaluating each combination on the FEVER score. Hou et al. [15] proposed a prompt-based approach for few-shot table-related tasks fact verification, utilizing manually designed templates to guide pre-trained models. Shi et al. [14] studied cross-language and cross-domain rumor detection, highlighting its importance in maintaining a healthy social media environment. They proposed a method superior to existing benchmark models, which effectively adapts from the source language and domain to the target by disentangling the semantic and emotional features of the dataset. Phan et al. [20] conduct a comprehensive survey of graph neural network (GNN) methods for fake news detection. Their work reviews all searchable articles on this topic, providing a detailed overview and categorisation of GNN models used in fake news detection systems. Lee et al. [11] introduced a innovative method that employs perplexity scores derived from PLMs to alleviate the data scarcity problem. Li et al. [21] introduce a novel Evidence-Providing Rumor Detection (EPRD) model aimed at not only detecting rumors on social media but also providing explanations to refute them. The EPRD model retrieves evidence from prior knowledge sources and examines the relationship between the evidence and the suspicious statement. It then constructs two heterogeneous graph objects to simulate the propagation of comments and evidence relationships. Utilizing the GraphSAGE component and an attention mechanism, EPRD enhances rumor detection accuracy.

B. Datasets

The efforts of datasets collection in this research field is also progressing significantly. Pan et al. [13] conduct a thorough empirical investigation across the few-shot generalization for fact verification, and thus construct a benchmark collection to gain a robuster model from well-resourced domains that generalized well on low-resource domains. Murayama et al. [22] conducts a comprehensive survey of 118 datasets related to fake news research, categorised into fake news detection, fact verification, and other tasks such as satire detection. They also details the utilisation tasks and characteristics of these datasets and discusses the challenges in constructing fake news datasets and potential research opportunities. Thorne et al. [23] introduce the FEVER dataset, which includes 185,445 claims derived from Wikipedia, and verified by annotators. They highlight the challenges of claim verification, particularly the need for many-hop reasoning and evidence extraction from multiple sources. Augenstein et al. [24] introduce MultiFC, the largest publicly available dataset for automatic claim verification, analysing its characteristics and challenges, and demonstrating its significant performance improvements. Jiang et al. [25] introduce HoVer (HOppy VERification), a dataset designed to challenge models with many-hop evidence extraction and fact verification from multiple Wikipedia articles. D'Ulizia et al. [26] systematically review twenty-seven popular FEND datasets, providing insights and a comparative analysis of their characteristics to guide researchers in selecting suitable datasets for evaluating FEND methods. Nørregaard et al. [27] present a dataset, DanFEVER, intended for multilingual misinformation research. DanFEVER is in Danish and has the same format as the well-known English FEVER dataset for testing methods in multilingual settings.

It is noteworthy that the above discussed solutions exhibit a significant limitation: they focus solely on syntactic- or semantic-level interrelationship and model architecture-level optimization without enhancements on supervisory data [5], [28]. In this work, we brought Det2Ver, which leverages the external knowledge from rumor detection as an additional supervisory signal for few-shot fact verification.

III. Rumor Detection to Verification (Det2Ver)

This work is motivated by the linguistic and semantic consistency between rumor detection and verification. We observed that these two tasks are intrinsically linked, yet researchers typically separate them, developing distinct processing models for each. Rumor detection aims to assess the truthfulness of news/hypotheses directly, whereas fact verification evaluates their authenticity based on given premises.

Exploring their unification strategies to exert the potentials of cross-task synergies seems to be feasible.

For instance, given a rumor [HYPOTHESIS]: "`Eating carrots can significantly improve eyesight because they contain a lot of vitamin A.`", rumor detection identifies that it's fake. On this basis, given a premise [PREMISE]: "`While vitamin A deficiency can lead to vision problems, most people can obtain sufficient vitamin A through a balanced diet.`", fact verification can also recognize it's fake based on this evidence. Intuitively, if we suspend the [HYPOTHESIS] to the [PREMISE] using prompting templates, we can get a new type of rumor: "`Based on [premise/evidence], we can assume it is true that [hypothesis/claim].`", which can also be treated as a novel rumor detection task instance.

Considering the rapid development of Generative LLM-based prompt-tuning learning, their amalgamation could lead to extraordinary, innovative outcomes. Thus, we focuses on integrating the two tasks into same form by using prompt-learning, and proposed a PLMs-based prompt-tuning framework for few-shot fact verification, Rumor Detection to Fact Verification (Det2Ver). By leveraging the rich, high-quality supervisory information from the former, transferred knowledge thus can aid and enhance the latter task's performance under data-scarce scenarios. Det2Ver belongs to cross-task knowledge transfer learning, in which the first step should be the training and inference unification of the two tasks: rumor detection and fact verification.

As depicted in Figure 2, following the scheme described in prompt learning survey [29], Det2Ver is divided into three parts:

1) The devised *prompting consolidation mechanism* during prompt engineering: in this stage, the input for the fact verification task, which includes the evidence and the claim, is combined using a prompting template and organised into the same single-sequence format as rumour detection, then encoded by the pre-trained language model (PLM);

2) The *parameter efficient fine-tuning* at PLMs reasoning/tuning stage: in this stage, the generative language model is used to encode the prompt and perform sequence generation decoding operations;

3) The *label words synchronization* during answer engineering: in this stage, the classification labels for both tasks are mapped and unified based on the predictions of the language model, i.e., the output probability distribution. This completes the formal unification of the two tasks, their knowledge sharing, and cross-task knowledge transfer.

A. Prompting Consolidation Mechanism

The uniformity in the input format of rumor detection and fact verification is a prerequisite for cross-task knowledge transfer. Specifically, we can adjust the input by utilizing prompting templates to ensure cross-task compatibility and adaptation. Since dimensionality reduction is easier than increasing dimensions and training a binary classification model is more stable, we chose to reduce the dimensions of the three-category fact verification task to fit the binary classification model for rumour detection in our experiments.

On the basis that Det2Ver is a binary classification model, we integrate the inputs of two tasks to make them compatible with Det2Ver. As shown in Figure 2 left, specifically, we maintain the original input of the rumor detection unchanged, and for the downward compatibility of fact verification to rumor detection tasks, we achieve this through hierarchical transformation in the prompting templates, which are shown in Appendix A and Table VII. This transformation comprises a two-layer prompting operation: the *internal prefix prompting* (Three prompting variants {true/uncertain/false} which helps in the label dimension decomposition and integration between the binary detection classification and the tertiary verification classification) and the *consolidation prompting* (Nine consolidation prompts which ensure the input compatibility from fact verification to rumor detection), integrating the hypothesis/claim with the premise.

During the training/prediction phases, given the original inputs for this two tasks, `Det_In` and `Ver_In`:

- For rumor detection, the input for Det2Ver remains the original input `Det_In` unchanged;
- For fact verification, the input for Det2Ver consists of three prompting variants combined with different internal prefixes and random selected consolidation prompts.

To gain a better understand of how the prompting consolidation mechanism works, we also provide an example of data feeding transformation, as depicted in Appendix B and Table VIII.

B. Parameter Efficient Fine-tuning

Mainstream prompt-tuning techniques usually optimize all trainable parameters of the PLM, which is computationally intensive. This approach can often lead to significant performance fluctuations in few-shot training. Thus, we choose to adopt the parameter efficient fine-tuning (PEFT) technique to train Det2Ver, specifically the Low-Rank Adaptation (LoRA) [30].

As shown in the mid of Figure 2, we implemented the LoRA LM-tuning strategy based on the initial code by T-Few[1], focusing on optimizing the rank decomposition matrices (low-rank matrices) of the dense layers during adaptation. By doing so, we not only preserve the integrity of the PLM but also achieve better performance with a relatively small number of added parameters, all while keeping the PLM's parameter weights unchanged.

Following the setting of Inhibiting and Amplifying Inner Activations (IA)[3] from T-Few [31], we use the following three loss function as the optimization objectives for Det2Ver to update the additional learnable parameters.

- **Conditional Language Modeling Objective**: As for probabilistic language modeling, we should maximize the rationality of correct intermediate answers. Formally, given the input sequence X and the correct target sequence $Y = (y_1, y_2, ..., y_T)$, we use the cross-entropy

[1]T-Few Code: https://github.com/r-three/t-few.

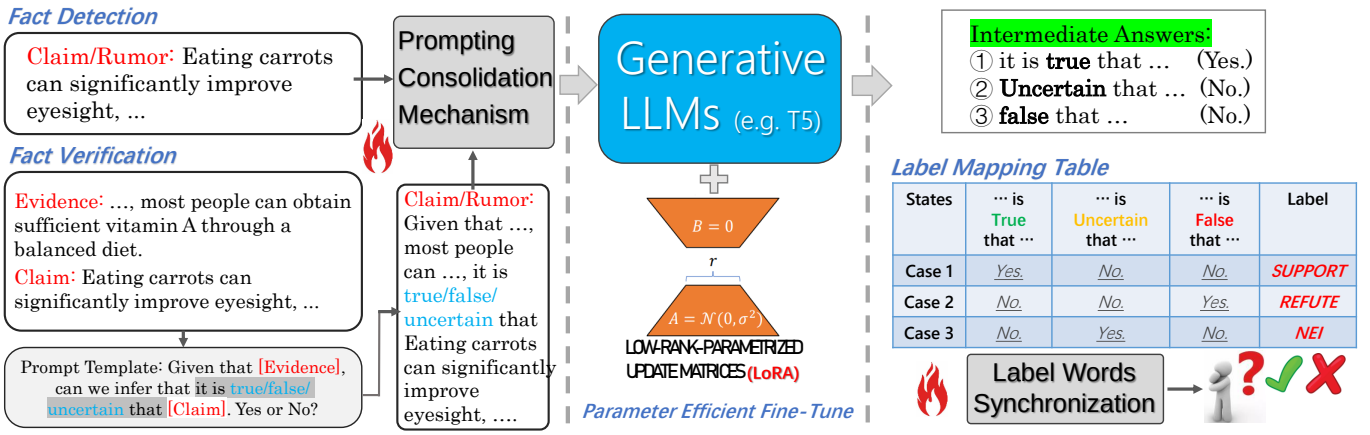


Fig. 2. The overall architecture of our Det2Ver framework, which we have split into three primary stages. The first stage (left side of the figure) is about the self-adaptive unification of the rumour detection and fact verification tasks by using prompt construction. The second stage (middle of the figure) involves Parameter Efficient Fine-tuning, where the generative language model is used to encode the prompt and perform sequence generation decoding operations. The third stage (right side of the figure) employs the label words synchronization engine. Based on the predictions of the language model, i.e., the output probability distribution, the classification labels for both tasks are mapped and unified.

loss L_{LM} in Equation1 to train Det2Ver to increase the probability of the correct target sequence;

- **Unlikelihood Loss**: We evaluate using rank classification, which assigns labels based on the ranking of probabilities for correct and incorrect sequences. Consequently, we aim to minimize the likelihood of incorrect target sequences. This approach is intended to discourage Det2Ver from predicting incorrect sequences.
  Given a set of N incorrect target instances, with T^i representing the sequence length of the i-th instance, the likelihood loss, denoted as L_{UK}, is comprehensively detailed in Equation 2.
- **Length Normalization Loss**: Given the impact of target sequence length on performance, as in our experiment where the candidate answers are {Yes., No.}, we incorporate an additional loss term in training for length-normalized evaluation. We first calculate the length-normalized log probability of an output sequence. Then, by minimizing the softmax cross-entropy loss, we maximize the length-normalized log probability of the correct answer. As illustrated in Equation 3, the term $\delta(\mathbf{X}, \mathbf{Y})$ represents the length-normalized log probability of generating a target sequence Y from a given input sequence X (the reciprocal of L_{LM}).

$$L_{\mathrm{LM}} = -\frac{1}{T}\sum_{t} \log p\left(y_t \mid \mathbf{X}, y_{<t}\right), \tag{1}$$

In Equation 1, the L_{LM} is a standard cross-entropy loss used to calculate the joint conditional probability of the output sequence given the original input X. This loss function encourages the adopted language model LM to assign a higher probability to the correct target sequence. In this formula, X denotes the original input to the language model. $p\left(y_t \mid \mathbf{X}, y_{<t}\right)$ represents the conditional probability of the token y_t given the input sequence X and all previous tokens $y_{<t}$. T is the length of the output sequence from the language model.

$$L_{\mathrm{UK}} = -\frac{\sum_{i=1}^{N}\sum_{t=1}^{T^{(i)}} \log\left(1 - p\left(\hat{y}_t^{(i)} \mid \mathbf{x}, \hat{y}_{<t}^{(i)}\right)\right)}{\sum_{i=1}^{N} T^{(i)}}, \tag{2}$$

In Equation 2, L_{UK} is designed to guide the language model L_{LM} to minimize the probability of incorrect target sequences, encouraging the model to reduce the likelihood of generating these sequences. In this formula, $\sum_{i=1}^{N} T^{(i)}$ is the number of incorrect target instances. $T^{(i)}$ is the sequence length of the i-th instance. $p\left(\hat{y}_t^{(i)} \mid \mathbf{x}, \hat{y}_{<t}^{(i)}\right)$ denotes the probability of the t-th token of the i-th incorrect target sequence.

$$L_{\mathrm{LN}} = -\log(\exp(\delta(\mathbf{X}, \mathbf{Y}))/ \left(\exp(\delta(\mathbf{X}, \mathbf{Y})) + \sum_{n=1}^{N} \exp\left(\delta\left(\mathbf{X}, \hat{\mathbf{Y}}^{(n)}\right)\right)\right), \tag{3}$$

In Equation 3, to account for the impact of target sequence length, L_{LN} normalize the log probability of an output sequence and maximize this normalized log probability for the correct answer. In this formula, $\delta(\mathbf{X}, \mathbf{Y})$ represents the length-normalized log probability of generating the target sequence Y from the input sequence X. $\hat{\mathbf{Y}}^{(n)}$ is the n-th incorrect target sequence

As shown in Equation 4, we simply sum the losses, $L_{\mathrm{LM}}, L_{\mathrm{UK}}, L_{\mathrm{LN}}$, following the settings of T-Few [31] and ProToCo [9], which avoids the debugging complexity of few-shot learning.

$$L_{total} = L_{\mathrm{LM}} + L_{\mathrm{UK}} + L_{\mathrm{LN}}, \tag{4}$$

To better illustrate the chain of information, we have added a running example corresponding to this optimization step in Appendix C, Practical Implementation Example Analysis. (refer to the Det2Ver framework in Figure 2 for further explanation).

C. Label Words Synchronization Engine

The output labels of Det2Ver, within a binary classification space {Yes, No}, are not suitable for direct use in fact verification. To address this, we developed a label-word synchronization mechanism to transform intermediate outputs into verification-ready labels, involving three classes: {SUPPORT, REFUTE, and NOT ENOUGH INFORMATION (NEI)}. In essence, the inspiration for this mechanism comes from the need to harmonize different classification outputs and improve the model's ability to generalize and perform well in both tasks by leveraging the available data more effectively. This innovative engine ensures that the model can robustly handle various types of misinformation, enhancing overall performance and reliability. The core idea is to create a unified label space that can accommodate the outputs from both tasks. This allows the model to leverage the strengths and data from one task to improve the performance in the other.

This mechanism/engine comprises two stages: *lookup table-based retrieval* and *target sequences probability ranking*.

In the first stage, *lookup table-based retrieval*, we utilize a label mapping table to align the LM probabilistic distribution with candidate answers, employing *Cross-Entropy* for precise binary classification Yes, No determination. As shown in Figure 2, this enables mapping and retrieval of final labels from the table for designated target states.

For cases not covered in the mapping table, the second stage involves *target sequences probability ranking*, which employs a ranking system based on the probabilistic alignment of target states with LM-generated probabilities, using *Cross-Entropy* as a metric. The process involves averaging and ranking these probabilities, selecting the candidate state that yields the minimal average cross-entropy, thus optimizing the label selection for fact verification.

Here, we provide the detailed methodologies and case studies of Label Words Synchronization.

For a given verification instance pair premise, claim, we firstly create three prompting variants corresponding to {true/uncertain/false}. Subsequently, we obtain the language modelling probability distribution D_{LM} from LM under each prompt, as shown in Equation 5.

$$D_{\mathrm{LM}} = \{p\left(y_t \mid \mathbf{X}, y_{<t}\right)\}_t^T, \tag{5}$$

where the $p\left(y_t \mid \mathbf{X}, y_{<t}\right)$ denotes conditional probability of the token y_t given the input sequence X and all previous tokens; T is the length of the target sequence generated by the language model. This is crucial for understanding how well the language model predicts each token in the sequence, which in turn affects the overall performance of the rumor detection and fact verification tasks.

Lookup Table-based Retrieval: Then, we calculate the length-normalized cross-entropy $Logit_a$ between the language probability distribution D_{LM} and answer $a = \{w_i^a\}_1^T$ in candidate answer choices $A = \{Yes, No\}$, which can be considered as the confidence level of target predictions. We denoted it as follows.

$$Logit_a = \frac{\sum_{i=1}^{T}\{CE(D_{\mathrm{LM}}, w_i^a)\}_{i=1}^{T}}{T}, \tag{6}$$

where $CE(\cdot)$ is the CrossEntropy operation, w_i and T denote the i-th token in candidate answer and the length of candidate answer.

Then, we consider the candidate answer $\hat{a}$ with the minimum $Logit_a$ as the final prediction result, as shown in Formula 7.

$$\hat{a} = \underset{\hat{a}}{Argmax}(-Logit_{\hat{a}}), \tag{7}$$

Finally, as depicted in Table I, using this label mapping table, we match the intermediate results to the verification label in {SUPPORT, REFUTE, NEI}.

TABLE I
INTERMEDIATE ANSWER MAPPING TABLE IN FIGURE 2.

Label	...is true that ...	...is uncertain that ...	...is false that ...
SUPPORT	Yes	No	No
REFUTE	No	No	Yes
NEI	No	Yes	No

Sequences Probability Ranking: For cases not covered in the answer mapping table I, we determine the verification label by assessing the fit ranking between the LM output probabilities and the decomposition sub-labels corresponding to fact verification.

Specifically, given S ={SUPPORT, REFUTE, NEI} and $s \in S$, we firstly obtain the set D by mapping s to candidate answer choices $A = \{Yes, No\}$ through the mapping table I. For example, if $s = SUPPORT$, then $D_s = \{Yes, No, No\}$. Given the intermediate answer $d \in D_s$ and the LM output probability distribution $D_{LM}^d \in \{D_{LM}^{true}, D_{LM}^{uncertain}, D_{LM}^{false}\}$ of the prompt variant corresponding to d, we calculate the length-normalized cross-entropy $Logit_{d|s}$ between d and D_{LM}^d, which is similar to Equation 6.

$$\hat{s} = \underset{\hat{s}}{Argmax} \sum_{d \in D_{\hat{s}}} (-Logit_{d|\hat{s}}). \tag{8}$$

Finally, as shown in Equation 8, we choose the top-ranked label $\hat{s}$ in {SUPPORT, REFUTE, NEI} that minimize the sum of $Logit_{d|\hat{s}}$ of three prompting variants as the final result for fact verification.

D. Leveraging External Knowledge from Heterogeneous Tasks

The proposed Det2Ver essentially operates as a framework for knowledge transfer across heterogeneous tasks, fully leveraging external knowledge derived from rumor detection to enhance fact verification. Here, we provide a further Methodology Summarization of Det2Ver, emphasizing the implementation process of leveraging external knowledge from rumor detection. This process is achieved through the following steps:

- **Unified Task Input Format:** In rumor detection tasks, the input is typically a straightforward text classification task with labels such as "true" or "false." In fact verification tasks, the input includes a claim and corresponding evidence, with labels such as "support," "refute," or "not enough information." Det2Ver unifies the input formats

of rumor detection and fact verification by constructing unified prompt templates. Specifically, Det2Ver employs internal prefix prompts (e.g., "true/uncertain/false") and consolidation prompts (e.g., "Given [evidence], we can infer that [claim] is true/false/uncertain") to standardize the input format.

- **Parameter-Efficient Fine-Tuning:** During the model training phase, Det2Ver utilizes parameter-efficient fine-tuning techniques (such as LoRA) that optimize only a subset of parameters without altering the entire pre-trained language model's parameters. This method reduces computational costs while retaining the model's original capabilities, allowing it to be fine-tuned based on knowledge from rumor detection tasks, thereby improving its performance in fact verification tasks.
- **Label Synchronization Engine:** To enable knowledge sharing between rumor detection and fact verification tasks, Det2Ver has developed a label synchronization engine. This engine aligns the language model's probability outputs with candidate answers through cross-entropy calculations and converts intermediate results into final fact verification labels (support/refute/not enough information) based on a label mapping table. This method ensures that the output labels of both tasks can be compared and mapped within the same space, facilitating cross-task knowledge transfer.
- **Cross-Task Knowledge Transfer:** Based on the above steps, Det2Ver leverages the rich supervisory signals from rumor detection tasks and transfers this knowledge to fact verification tasks through unified prompt templates and the label synchronization engine. This cross-task knowledge transfer not only enhances the model's generalization capabilities in low-resource scenarios but also improves its overall performance in fact verification tasks.

By employing these methods, Det2Ver effectively transforms external knowledge from rumor detection tasks into useful information for fact verification tasks, significantly enhancing the model's performance in handling complex fact verification tasks.

IV. Experimental Settings

A. Overview

In terms of datasets, we utilized three public datasets: FEVER [23], VitaminC [7], and SciFACT [32]. We incorporated the following external supervised corpora to Det2Ver: LIAR [33], FakeNewsNet [34], and COVID-19 Fake News [35]. To benchmark against baselines, we chose RoBERTa-L [11], GPT2-PPL[2], SEED [12], and ProToCo[3] [9], all of which are latest models proposed for few-shot fact verification. The adopted evaluation metric is Macro-F1.

B. DataSets

Regarding the rationale behind choosing the mentioned datasets for the experiments section, we primarily considered the datasets' general applicability, official accuracy, and overall annotation quality. Many publicly available datasets lack systematic and standardised annotation procedures. For example, the selection of FEVER, as discussed in Murayama's investigation [22], was motivated by its comprehensive nature. The FEVER dataset for the first FEVER shared task consists of plenty of claims, generated by altering sentences extracted from Wikipedia and pre-processed Wikipedia data. Furthermore, the FEVER dataset is officially used in these forums or competitions promoting fact-verification research, making it a reliable and well-recognised benchmark for comparative studies.

The specific descriptions of the three datasets used for the fact verification task, FEVER, VitaminC, and SciFACT, are as follows:

FEVER[4] [23] is an open-domain dataset that contains over 185,000 pairs of human-crafted factual verification descriptions, all sourced from introductory pages on Wikipedia. However, different from [23], the downloaded version[5] does not include the '`Not Enough Info`' (`NEI`) category. To address this, we utilize an revised version of FEVER provided by [9], which constructs `NEI` instances by uniform sampling unrelated claims and premises.

VitaminC[6] [7] is an robust adversarial fact verification dataset with contrastive premise, particularly its flexible sensitivity to subtle information changes. VitaminC have a contrastive nature: each evidence pair is nearly identical in language and content, yet one supports the given claim while the other contradicts it. It assembles over 100,000 Wikipedia revisions that modify underlying facts, along with synthetically generated data, culminating in over 400,000 claim-evidence pairs.

SciFACT[7] [32] is a scientific domain-based claim verification dataset, deriving evidence or premises from abstracts in research literature to support or refute a given scientific claim. It comprises 1,409 scientific claims, each accompanied by expert-annotated abstracts that either support or refute the claim, and includes rationales justifying each decision. It is much smaller in scale compared to FEVER and VitaminC. For instance, the total of `Support/NEI/Refute` claims is only about one percent of FEVER, To maintain comparison consistency, we also choose the revised SciFACT provided by [9].

Details on the training/testing split of instances, as well as the category ratios for the fact verification datasets, are provided in Table II.

The rumor detection datasets used for the cross-task knowledge transfer of Det2Ver include: LIAR[8], FakeNewsNet[9], and COVID-19 Fake News[10].

For rumor detection datasets, we applied criteria similar to those used for fact verification datasets. When selecting

[2]GPT2-PPL: https://github.com/HLTCHKUST/Perplexity-FactChecking.

[3]ProToCo: https://github.com/znhy1024/ProToCo.

[4]FEVER: https://github.com/awslabs/fever.

[5]FEVER: http://fever.ai/.

[6]VitaminC Dataset: https://github.com/TalSchuster/VitaminC.

[7]SciFACT Dataset: https://github.com/allenai/scifact.

[8]LIAR:https://github.com/thiagorainmaker77/liar_dataset.

[9]FakeNewsNet Dataset: https://github.com/KaiDMML/FakeNewsNet.

[10]**COVID-19 Fake News**: https://github.com/parthpatwa/covid19-fake-news-dectection.

TABLE II
STATISTICS OF THE ADOPTED FACT CHECKING DATASETS.

Dataset		SUPPORT	NEI	REFUTE	TOTAL
FEVER	Train	80035	35517	29775	145327
	Test	3333	3319	3333	9985
VitaminC	Train	124864	52981	71108	248953
	Test	17306	7268	9907	34481
SciFACT	Train	332	304	173	809
	Test	124	112	64	300

LIAR [33], FakeNewsNet [34], and COVID-19 Fake News [35] datasets, especially LIAR, we focused on the datasets' scale, broad applicability, and overall annotation quality. These factors are crucial because well-annotated datasets enable more accurate and meaningful experiments. As highlighted in Murayama's survey, the LIAR dataset, a large-scale resource in fake news detection, automatically extracts justifications provided in fact-checking articles associated with each claim.

LIAR [33] is a decade-long dataset for automatic rumor detection, which consists of 12.8K manually labeled short statements collected from POLITIFACT.COM[11].

LIAR, containing genuine, real-world brief statements, surpasses previous datasets in size by an order of magnitude, facilitating the creation of statistical and computational methods for detecting fake news. It is divided into training, validation, and test sets, and includes diverse contexts like news releases, interviews, campaign speeches, and social media posts. Specifically, LIAR is divided into three parts: the training set consists of 10,269 statements, the validation set contains 1,284 statements, and the test set has 1,283 statements. The average statement length is 17.9 tokens. The dataset features top speaker affiliations with 4,150 statements from Democrats, 5,687 from Republicans, and 2,185 from sources with no specific affiliation, such as Facebook posts.

FakeNewsNet [34] is a multidimensiona data repository for fake news detection, containing a considerable volume of corpora with diverse features in news and social context, and spatiotemporal information.

FakeNewsNet, developed by Shu et al. [34], includes two main sources: PolitiFact and GossipCop, focusing on political and entertainment news, respectively. PolitiFact contains 432 fake and 624 real news articles, labeled by fact-checking experts. And GossipCop includes 5,323 fake and 16,817 real news articles, with similar expert annotations. Based on this, FakeNewsNet is split with PolitiFact containing 432 fake and 624 real news articles, and GossipCop including 5,323 fake and 16,817 real news articles. FakeNewsNet dataset supports fake news detection, propagation, and mitigation, providing essential tools for combating misinformation on social media.

COVID-19 Fake News [35] is a manually annotated dataset of over 10 thousands social media articles and posts of COVID-19 real/fake news.

[11]**PolitiFact WebSite**: http://www.politifact.com/.

COVID-19 Fake News[12] is presented to combat the spread of misinformation related to COVID-19. It comprises 10,700 social media posts and articles, which are manually annotated as either real or fake news. The data was collected from various platforms including Twitter, and fact-checking websites such as Politifact and Boomlive.COVID-19 Fake News includes English-language textual content related to COVID-19. Fake news was identified using public fact-verification websites, while real news was gathered from verified Twitter handles of credible sources like WHO and CDC. COVID-19 Fake News is balanced with 52.34% real news and 47.66% fake news, and it is split into training (60%), validation (20%), and test (20%) sets.

C. Baselines

We leverage the following several SOTA few-shot fact verification approaches as comparison baselines: Majority, RoBERTa-L, GPT2-PPL[13] [11], SEED [12] and ProToCo [9].

Majority, firstly introduced by [11], is a straightforward way that assigns the most frequently occurring label in training data to all testing data. It is often regarded as a direct and intuitive strategy, especially in the context of imbalanced category distribution in the training set.

RoBERTa-L, which employs a feed-forward network as a straightforward downstream classifier for the unidirectional LM, RoBERTa-Large [36], is a commonly used approach to achieve SOTA performance in the literature;

GPT2-PPL, utilizing a single-parameter classifier based on the perplexity score by the generative LM, GPT2-base [37], stands as a SOTA few-shot solution. Unlike traditional fine-tuned classifiers, this model is distinguished by its '-PPL' suffix, indicating its reliance on perplexity-based classification. This unique approach enables it to surpass the performance of models such as BERT, RoBERTa, and XLNet.

SEED, an acronym for Semantic Embedding Element-wise Difference, represents a novel vector-based approach for few-shot claim verification. This approach aggregates pairwise semantic differences for claim-evidence pairs. The average semantic differences captured for claim-evidence pairs by class representative vectors are subsequently utilized in rumor/claim classification.

ProToCo is a SOTA prompt-based method for few-shot fact assessment, incorporating a factuality-grounded consistency constraint. By learning from self-consistent variant inputs. It further enhances the semantic consistency between the categories, {SUPPORT, REFUTE, NEI}.

D. Evaluation Metric

In this study, we utilize Macro-F1 as the evaluation metric. Macro-F1 is a commonly used metric for evaluating the performance of classification models.

Its calculation is based on: Precision and Recall. The Macro-F1 score is particularly useful in scenarios

[12]The data of COVID-19 Fake News are available at https://github.com/parthpatwa/covid19-fake-news-dectection.

[13]Both the implementations of RoBERTa-L and GPT2-PPL are used from: https://github.com/HLTCHKUST/Perplexity-FactChecking.

with imbalanced category distribution. Mathematically, the `Macro-F1` score is defined as follows: Firstly, compute the `Precision` and `Recall` for each class through:

$$\text{Precision}_i = \frac{\text{True Positives}_i}{\text{True Positives}_i + \text{False Positives}_i} \tag{9}$$

$$\text{Recall}_i = \frac{\text{True Positives}_i}{\text{True Positives}_i + \text{False Negatives}_i} \tag{10}$$

Then, compute the `F1` score for each class through:

$$F1_i = \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{11}$$

Finally, calculate the `Macro-F1` score by taking the average of the $F1_i$ scores across all classes:

$$\text{Macro-F1} = \frac{1}{N} \sum_{i=1}^{N} F1_i \tag{12}$$

where N is the number of classes.

This metric gives equal importance to all categories, preventing categories with more samples from dominating the overall performance evaluation. Therefore, this metric is very suitable for the dataset applied in our experiments.

E. Hyperparameters

Here, we provide the experimental settings of Det2Ver, such as learning rate, batch size, and training epochs. We conducted a comprehensive grid search to identify the optimal hyperparameters for Det2Ver, ensuring peak performance. We set the learning rate at 0.0001 and employed a scheduler for linear decay complemented by a warm-up phase. The training was structured to extend up to a maximum of 1500 steps, incorporating testing evaluations at intervals of every 50 steps. The training and inference processes were designed with a maximum input token length limited to 250 tokens. To ensure efficiency, we implemented a patience mechanism of 5, meaning that the training would be preemptively halted if no performance improvements were observed after five consecutive evaluations. To avoid out-of-memory (OOM) issues, the batch size during the training phase was maintained at 1. For the inference stage, the batch size was increased to 4 to optimize processing efficiency. For the Det2Ver backbone, we selected the versatile multitask prompt-tuned T5 (T0) [38], [31], an advanced generative LLM.

In terms of few-shot learning, we configured $K = \{4, 8, 16, 32\}$ for K-Shot. We randomly sample {4, 8, 16, 32} instances from the adopted fact verification datasets. From each rumor dataset, we randomly selected $\{10, 25, 50\}$ instances as cross-task external supervised knowledge to enhance Det2Ver's verification performance.

For baselines, such as RoBERTa-L, GPT2-PPL [11], and T-Few [31], we replicated their performance metrics using previously established settings as detailed in existing literature. Meanwhile, the results for models like SEED [12] and ProToCo [9] were directly sourced from their respective publications. All experiments above were conducted on 4 $\times$ NVIDIA RTX A6000 48GB GPUs.

V. Experiments

A. Main Results

To evaluate the effectiveness of Det2Ver, we conducted comprehensive few-shot learning experiments with $K = \{4, 8, 16, 32\}$ shots on three public datasets, **FEVER**, **VitaminC** and **SciFACT**. As shown in Table III, the first group of results represents the performance of selected comparative baseline models. The second group of results demonstrates the performance of our proposed Det2Ver under various configurations: Det2Ver(20), Det2Ver(50), and Det2Ver(100), where the numbers in parentheses indicate the amount of training examples incorporated from each of the three detection datasets: **FakeNewsNet**, **LIAR** and **COVID-19 Fake News**. We tested the impact of integrating different numbers of rumor detection training examples {20, 50, 100} into the training process and analyzed the consequent outcomes.

Baselines Summarization Firstly, the first group's results clearly indicates that ProToCo [9] is an exceptionally powerful model for few-shot fact verification. First, the results from the first group clearly indicate that ProToCo [9] is an exceptionally powerful model for few-shot fact verification. The factuality-grounded consistency constraint introduced by ProToCo greatly enhances its generalization capabilities, even with a few supervised data, with T-Few [31] closely following in performance. Both models notably surpass other methods, including SEED and GPT2-PPL, due to the robust T0 they are built upon [38]. T-Few and ProToCo, being based on T0, leveraging T0's capabilities, benefit from multi-task supervised learning and extensive mixed training across a wide range of NLP tasks. This has significantly contributed to the enhanced generalizability of language models in new NLP tasks. Therefore, this became the primary motivation for us to proactively choose the T0 (specifically T0-3B[14]) as the backbone of our Det2Ver.

Few-Shot Comparisons to SOTA Baselines Secondly, comparing the data in Table III, Det2Ver has surpassed T-Few and, in some cases, outperformed ProToCo with only 20 additional examples from a rumor detection dataset. On the FEVER dataset under 4-shot setting, it achieved an absolute 0.8% Macro-F1 improvement, and in the 32-shot setting, it recorded gains of 1.5% and 2.6% in Macro-F1 on the VitaminC and SciFact datasets, respectively. The second group recorded the best and second-best performances among all models. Notably, Det2Ver model variants, Det2Ver(50) and Det2Ver(100), lead significantly in Macro-F1 performance compared to other baselines, with increased external supervisory signals from rumor detection. For instance, we would like to emphasize that our Det2Ver consistently surpasses the state-of-the-art ProToCo by approximately 5% to 10% absolute gains in Macro-F1 across the majority of few-shot settings and datasets. This indicates that rumor detection can effectively serve as external knowledge for fact verification, enhancing its generalizability and performance, thereby ensuring the effectiveness of Det2Ver for fact verification data-scarce scenarios.

Quantitative Analysis of Detection Data Influence We evaluated and discussed the impacts of different amounts of

[14]T0-3B:https://huggingface.co/bigscience/T0_3B.

TABLE III
THE PERFORMANCE COMPARISONS BETWEEN OUR DET2VER AND OTHER FEW-/ZERO-SHOT FACT VERIFICATION BASELINES ON THREE PUBLIC BENCHMARKS, MEASURED USING MACRO-F1 SCORES, IN FEW SHOT $K = \{4, 8, 16, 32\}$ SETTINGS (IN WHICH WE SAMPLE K INSTANCES PER CATEGORY). TO ENSURE FAIR COMPARISONS, ALL REPORTED RESULTS REPRESENT THE MEAN VALUES OBTAINED FROM FIVE INDEPENDENT TRIALS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, WITH THE SECOND-BEST SCORES UNDERLINED FOR BOTH BASELINE GROUPS AND OUR DET2VER. NOTABLY, GREEN AND BLACK NUMBERS INDICATE THE RESPECTIVE INCREASES AND DECREASES IN THE MARGINS BETWEEN OUR DET2VER'S PERFORMANCE AND THE BEST BASELINE RESULTS.

Few-Shot Settings	**# 4-Shot**			**# 8-Shot**			**# 16-Shot**			**# 32-Shot**		
Models (% Macro-F1)	FEVER	VC	SciFACT	FEVER	VC	SciFACT	FEVER	VC	SciFACT	FEVER	VC	SciFACT
Majority	16.7	22.3	19.5	16.7	22.3	19.5	16.7	22.3	19.5	16.7	22.3	19.5
RoBERTa-L	16.9	14.6	21.0	19.3	17.4	29.7	27.7	24.3	37.4	32.3	42.1	47.2
GPT2-PPL	29.3	30.3	32.6	34.1	40.7	42.3	39.4	44.9	47.3	47.2	53.5	50.8
SEED	50.1	31.3	35.5	53.8	42.2	40.4	54.3	46.2	45.2	56.2	52.9	48.7
T-Few	85.1	48.9	38.2	87.6	54.5	44.2	88.1	56.8	54.5	90.6	59.7	58.3
ProToCo	**89.1**	**52.0**	**49.8**	**90.7**	**56.8**	**53.5**	**91.5**	**59.4**	**58.7**	**92.1**	**64.0**	**60.1**
Det2Ver(20)	89.9 (0.8 ↑)	50.2 (1.8↓)	45.4 (4.4 ↓)	90.2 (0.5 ↓)	53.4 (3.4 ↓)	55.4 (2.1 ↑)	91.0 (0.5 ↓)	55.2 (4.2 ↓)	58.3 (0.4 ↓)	91.3 (0.8 ↓)	65.5 (1.5 ↑)	62.7 (2.6 ↑)
Det2Ver(50)	90.1	54.4	52.3	91.0	56.8	58.7	91.7	61.8	67.6	**92.4**	67.4	69.6
Det2Ver(100)	**92.4**	**57.3**	**57.6**	**91.5**	**57.3**	**60.1**	**92.1**	**62.7**	**69.5**	**92.4**	**68.0**	**71.2**

supervised data, including both rumor detection and fact verification datasets. It is clear that additional rumor detection learning instances significantly enhance Det2Ver in performing fact verification. Table III also indicated that even a small quantity of data can lead to substantial performance improvements. For example, Det2Ver(20) performs quite well compared to T-few. It should be noted that, aside from incorporating cross-task joint training of Det2Ver, there is no major structural difference between Det2Ver and T-few. Additionally, we observed that on the three datasets, and across different few-shot settings, the improvement seen with Det2Ver(50) over Det2Ver(20) was notably greater than the improvement of Det2Ver(100) over Det2Ver(50). This suggests that a modest amount of rumor detection data as supervisory input is sufficiently effective, and the benefits of adding more rumor detection data gradually diminish.

The Statistical Significance of Detection Data To understand whether the observed synergies are statistically meaningful, we conducted statistical significance tests on the performance improvements. Using paired t-tests, we compared the Macro-F1 scores of Det2Ver(20) with T-Few and ProToCo. Paired t-tests are used to compare the means of two related samples, assessing the fluctuations in experiments. Here, it evaluates the impact of supervised data from different domains on the performance of the Det2Ver model. The results showed that the improvements were statistically significant with p-values <0.10, indicating that the additional rumor detection data indeed provides a meaningful enhancement to fact verification performance. Similarly, the improvements of Det2Ver(50) over Det2Ver(20) were also statistically significant, reinforcing the value of moderate increases in rumor detection data.

Performance Gains with Increasing Few-Shot Instances Moreover, we discovered that as the shot number in few-shot settings increases, the positive effect of the transferred supervision from rumor detection gradually decreases. For instance, under 8-shot setting, Det2Ver shows more significant improvements compared to ProToCo and T-Few than under a 32-shot settings, in which Det2Ver(100) obtains an Macro-F1 increase of 0.8% under 8-shot compared with *0.8%* Macro-F1 increase under 32-shot on FEVER. It showcases that the knowledge transferred from rumor detection tasks enables Det2Ver to effectively enhances its overall capability in fact verification, even in scenarios with minimal data availability. Building on the above, we draw the following conclusion: Empowered by knowledge acquired from rumor detection, Det2Ver exhibits a remarkable capacity for adeptly managing and excelling in fact verification tasks.

The Statistical Significance of Few-Shot Instances To further evaluate the statistical significance of performance gains with increasing few-shot instances, we also performed ANOVA tests across different shot settings. ANOVA (Analysis of Variance) is a statistical method used to compare the means of multiple groups to determine if there are significant differences between them. It helps to identify whether the differences between groups are greater than the variations within groups. Here, it is primarily used to test the interference fluctuations in inference performance caused by different number of supervised detection data. The results also indicated that the performance improvements were statistically significant, with p-values <0.05 for Det2Ver compared to other models in both 8-shot and 32-shot settings. This confirms that the observed improvements are not due to random variations and highlight the effectiveness of integrating rumor detection data for enhancing fact verification performance.

Connections and Differences Among Det2Ver and Previous Works Det2Ver leverages the inherent connections between rumor detection and fact verification. By utilizing

the supervisory signals from rumor detection, we enhance few-shot/zero-shot fact verification performance. Det2Ver integrates both tasks into a unified prompt-tuning framework, facilitating effective cross-task knowledge transfer, unlike traditional methods [10], [15] which treat them separately. Traditional rumor detection focuses on binary classification (real or fake), while fact verification involves three-way classification (support, refute, not enough information). Det2Ver bridges this gap with a unified approach. Moreover, unlike methods that rely on large-scale datasets, Det2Ver emphasizes few-shot and zero-shot learning, addressing data scarcity effectively. By incorporating supervisory signals from rumor detection, Det2Ver significantly improves the generalizability of fact verification models, unlike existing methods [12], [10], [15], [16] that focus mainly on model architecture optimization.

B. Zero-Shot Comparisons of Det2Ver

We conducted an zero-shot learning evaluation of Det2Ver and other baselines, with the results shown in Table IV. We found that among the baselines, ProToCo performed best overall on FEVER and VitaminC. Meanwhile, ProToCo and T-Few performs similarly on SciFACT.

TABLE IV
ZERO-SHOT COMPARISONS BETWEEN OUR PROPOSED DET2VER AND ICL, T-FEW, PROTOCO. EXPERIMENTAL FLUCTUATIONS (STANDARD DEVIATION) ARE INDICATED IN PARENTHESES. THE BEST RESULTS OF THE BASELINE ARE HIGHLIGHTED IN BOLD, AND COLORED TEXT IS USED TO REPRESENT THE ABSOLUTE PERFORMANCE COMPARISON OF DET2VER AGAINST THE BEST BASELINE RESULTS.

DataSet	Zero-Shot Setting				
	ICL (OPT-30B)	T-Few	ProToCo (T0-3B)	Det2Ver (20)	Det2Ver (50)
FEVER	0.347	0.425	**0.479**	0.500	0.512
	-	(0.01)	**(0.00)**	2.1% ↑	3.3% ↑
VitaminC	0.340	0.359	**0.386**	0.373	0.387
	-	(0.00)	**(0.00)**	1.3% ↓	0.01% ↑
SciFACT	0.332	**0.336**	0.331	0.366	0.374
	-	**(0.02)**	(0.02)	3.5% ↑	4.3% ↑

From this table, Det2Ver(20) achieved absolute Macro-F1 score improvements of 2.1% and 3.5% on FEVER and SciFACT, respectively. Compared to the best baseline results, Det2Ver(50) further improved the Macro-F1 score by approximate 3% to 4%. However, the overall performance of Det2Ver on the VitaminC dataset was relatively inferior. Det2Ver(20) performed slightly worse than ProToCo on VitaminC, while Det2Ver(50) paralleled ProToCo's performance. This phenomenon is primarily attributed to the distinct nature of VitaminC, which concentrates on biological nutritional science, in contrast to the coverage of general domains and social contexts of our utilized rumor detection datasets - LIAR, FakeNewsNet, and COVID-19 Fake News. The conspicuous divergence in domain characteristics between these external rumor detection datasets and VitaminC limits the knowledge transferability from the former general domains to the latter biological science contexts.

Furthermore, the construction of premise/ evidence pairs in VitaminC leverages a contrastive fact verification paradigm. Given this contrastive paradigm, we speculate that the supervisory knowledge from external rumor detection datasets, lacking prior adversarial training and recognition capabilities for contrastive contexts, fails to effectively help to distinguish contrastive instances.

In contrast, the superior performance of Det2Ver on FEVER and SciFact indicates that it effectively utilized the cross-task supervisory information from rumor detection tasks such as LIAR and FakeNewsNet. This highlights the importance of domain adaptability between rumor detection and fact verification tasks. In subsequent ablation experiments, we further discuss the impact of cross-tasks domain characteristic differences on the fact verification performance.

C. Ablation Studies

To in-depth explore the cross-task synergies between fact verification and rumor detection, we conducted comprehensive ablation studies, investigating the experimental impact of factors including data domain differences, Det2Ver itself's structural superiority, and the diversity of used consolidation templates.

Domain Divergence of Rumor Detection DataSets To examine the influence of different rumor detection datasets with varying domain characteristics on the fact verification performance, we systematically evaluated Det2Ver by controling and adjusting the proportion of data derived from rumor detection, as shown in Figure 3. Specifically, we conducted our evaluations under 4-shot setting to minimize the influence of the fact verification data. We benchmarked against Det2Ver(20), hereafter referred to as Det2Ver(AVG), which randomly selects 20 training instances from each of the three adopted rumor detection datasets. Meanwhile, we crafted three Det2Ver variants, each sourcing rumor detection data exclusively from one of the following: LIAR, COVID-19 Fake News, and FakeNewsNet, which correspond to Det2Ver(LIAR), Det2Ver(COVID), and Det2Ver(FNN), respectively. To ensure a fair comparison, each of the three variants utilized a total of 60 training instances, aligning with the volume of rumor detection data used in Det2Ver(AVG)[15].

From this table, we can observe that Det2Ver- (LIAR) significantly outperforms Det2Ver(AVG) across all three datasets. This improvement highlights the versatility, high-quality context, and robust cross-domain support of LIAR. In contrast, Det2Ver(COVID) shows slightly inferior performance on FEVER and VitaminC compared to Det2Ver(AVG), but excels on SciFACT. We speculate this is largely due to SciFACT predominantly featuring scenarios related to COVID-19 topics, thereby offering higher applicability for task transfer in this specific domain. However, Det2Ver(FNN) exhibits poor performance on all three fact verification datasets. Upon inspecting the FakeNewsNet dataset, we identified that its overall annotation quality is not up to the standard, which likely contributes to the diminished performance.

[15]To keep fair comparison, SciFact(20)+FEVER(20)+VitaminC(20)=60, thus, the model variants like Det2Ver(LIAR) leverages 60 training instances.

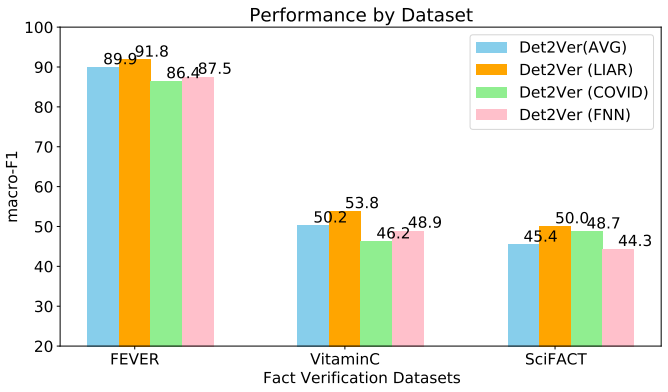


Fig. 3. The performance results on various types of fact verification datasets of Det2Ver variants that utilized different data proportions of rumor detection datasets.

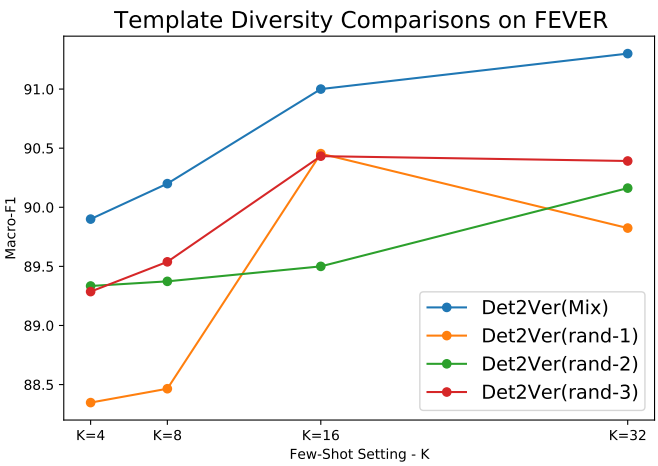


Fig. 4. Performance comparison of Det2Ver variants with diverse and individual consolidation prompting templates across different few-shot settings.

In summary, these observations indicate that the domain characteristics of different rumor detection datasets are significant factors influencing the performance of Det2Ver in fact verification tasks.

Det2Ver with No Rumor Detection Supervision To explore Det2Ver itself's architecture characteristics, we systematically evaluated Det2Ver without the assistance of external supervisory signals from rumor detection, denoted as Det2Ver(0), and compared with the baselines, T-Few, and ProToCo. Specifically, we conducted experiments on three datasets under both 4-shot and 16-shot settings, and the statistics are depicted in Table V.

TABLE V
COMPARISONS OF DET2VER IN FEW-SHOT SETTING WITHOUT RUMOR DETECTION DATA AGAINST SOTA BASELINES.

Models	# 4-Shot			# 16-Shot		
	FEVER	VC	SciFACT	FEVER	VC	SciFACT
T-Few	85.1	48.9	38.2	88.1	56.8	54.4
ProToCo	**89.1**	**52.0**	**49.8**	**91.5**	**59.4**	**58.7**
Det2-Ver(0)	87.6	44.6	40.3	88.1	52.0	57.6

As can be observed in Table V, compared to Det2Ver(0), which operates without any supervisory signals from rumor detection task, ProToCo remains the optimal model architecture relying solely on semantic-level consistency constraints in claim-evidence pairs and without external knowledge assistance. Furthermore, even in the absence of external supervision from rumor detection, Det2Ver exhibits a significant superiority over T-Few. This attributes its unique model architecture, particularly due to its various *Internal Prefixes* and *Consolidation Prompts*, which largely enrich the semantic-related training samples. The prompting consolidation mechanism of Det2Ver can also be regarded as an effective data augmentation strategy. Additionally, Det2Ver's binary output space and the *Label Words Synchronization Engine* further strengthen its few-shot learning power, enabling it to maintain certain advantages even without external supervisory information.

Diversity of Consolidation Prompting Templates Here, we explore the performance influence of the *Diversity of Consolidation Prompting Templates*. We employed up to 9 consolidation prompts to enhance the Det2Ver's generalizability during our experiments. As such, we use Det2Ver, which is augmented with all available consolidation templates, as a baseline comparison, denoted as Det2Ver(Mix). We also randomly selected three from the 9 templates, specifically templates T1, T4, and T5 in Table VII, Appendix A. And training three Det2Ver variants using only these individual templates, named Det2Ver(rand-1), Det2Ver(rand-2), and Det2Ver(rand-3), respectively.

Analyzing the experiment results depicted in Figure 4, it is evident that Det2Ver(Mix), which incorporates a diverse set of prompting templates, demonstrates superior performance compared to the variants that utilize only a single selected template. This suggests that leveraging a variety of templates to increase data diversity can significantly enhance Det2Ver's generalizability. The advantage of using multiple templates lies in their ability to introduce a broader range of linguistic structures and contexts, which helps Det2Ver learn more robust representations and improves its ability to generalize across different fact verification scenarios. The performance gains of Det2Ver(Mix) underscore the benefits of prompting template diversity, which can be a powerful strategy for improving the fact verification performance.

D. Case Studies

To understand the performance of Det2Ver in real-world scenarios and compare its effectiveness with existing methods, we included a series of user studies as follows.

Specifically, we selected several cases from these public datasets, FEVER, VitaminC, and SciFact, applying Det2Ver for analysis. More representatively, we choose Det2Ver(100) for comparisons, which involves random selection of 100 examples from each of the three rumor detection datasets: FakeNewsNet, LIAR, and COVID-19 Fake News, serving as external supervision corpus. Table VI showcases a subset

TABLE VI
CASE STUDY: FACT VERIFICATION PERFORMANCE OF DET2VER COMPARED TO THE BASELINES, SEED AND PROTOCO, ON FEVER, VITAMINC, AND SCIFACT DATASETS.

Claim / Hypotheses	Premises / Evidence	Gold Label	SEED	ProToCo	Det2Ver(100)
(FEVER) Jon Watts has only worked with Paramount Pictures.	Jon_Watts: Watts directed and co-wrote the upcoming Marvel Studios and Sony Pictures produced Spider-Man: Homecoming,slated for a July 7, 2017 release	REFUTE	REFUTE ✓	NEI ×	REFUTE ✓
(FEVER) Jane Eyre was written by Charlotte Bront.	Jane_Eyre: Jane Eyre (originally published as Jane Eyre: An Autobiography) is a novel by English writer Charlotte Bront.	SUPPORT	REFUTE ×	SUPPORT ✓	SUPPORT ✓
(VitaminC) By August 2015, WWE 2K15 shipped over 6 million units.	As of August 2015 , WWE 2K15 has shipped over 7.5 million units.	REFUTE **(tag error)** ->SUPPORT	NEI ×	REFUTE ×	SUPPORT ✓
(VitaminC) Bernard Sumner was credited as Barney Rubble by a New Order bootleg.	An early New Order bootleg credited him as ' Barney Rubble'.	REFUTE	REFUTE ✓	SUPPORT ×	REFUTE ✓
(SciFact) CX3CR1 on the Th2 cells suppresses airway inflammation.	CX3CR1 is required for airway inflammation by promoting T helper cell survival and maintenance in inflamed lung.	REFUTE	SUPPORT ×	NEI ×	REFUTE ✓
(SciFact) Deamination of cytidine to uridine on the minus strand of viral DNA results in catastrophic G-to-A mutations in the viral genome.	Broad antiretroviral defence by human APOBEC3G through lethal editing of nascent reverse transcripts: APOBEC3G is closely related to APOBEC1.	SUPPORT	REFUTE ×	SUPPORT ✓	SUPPORT ✓

of these cases, comparing the results of Det2Ver with other methods such as SEED and ProToCo.

In Table VI, the overall results show that Det2Ver(100) consistently aligns with the gold labels, reflecting its accuracy and reliability in fact verification tasks. For example, in the case of the claim about Jon Watts (the first row of group 1, FEVER dataset), Det2Ver correctly identifies the evidence that refutes the claim, which is consistent with the gold label. Similarly, for the claim about Jane Eyre (the second row of group 1), Det2Ver accurately supports the claim, aligning with the gold label. Det2Ver also demonstrates the ability to correct tagging errors in the data. For instance, the claim about WWE 2K15 (the first row of group 2, VitaminC dataset) was initially tagged as REFUTE due to an error. Det2Ver correctly identifies the supporting evidence, showing its robustness in handling data inconsistencies. Moreover, when compared to SEED and ProToCo, Det2Ver shows superior performance in several cases. For example, in the case of the claim that CX3CR1 on Th2 cells suppresses airway inflammation (first row of group 3, SciFact dataset), both SEED and ProToCo provide either SUPPORT or NEI (Not Enough Information), while Det2Ver accurately REFUTES the claim based on the given evidence.

These case studies highlight our proposed Det2Ver's capability to handle diverse and complex real-world scenarios, ranging from entertainment industry claims (FEVER dataset) to scientific statements (SciFact dataset). The Det2Ver framework's ability to leverage external knowledge from rumor detection tasks to enhance fact verification is evident from its consistent and accurate outputs across different domains.

VI. DISCUSSIONS

A. Method Characteristics in FSL and ZSL

Det2Ver in Few-Shot Learning (FSL): Det2Ver leverages the partially supervised datasets within the few-shot framework. Moreover, Det2Ver employs a unified framework that incorporates additional fake news detection datasets for auxiliary learning. This approach provides the model with more cognitive biases and guiding signals, enabling better generalization even with limited examples, i.e., few-shot learning. The cross-task knowledge transfer from fake news detection to fact-checking significantly enhances the model's performance in few-shot scenarios.

Det2Ver in Zero-Shot Learning (ZSL): Although no fact-checking task-specific datasets are used during ZSL training, Det2Ver adopts a T5-based prompt learning paradigm. By using well-designed prompt templates, the model acquires a certain zero-shot prediction capability. Additionally, the biases learned from the fake news detection datasets enable Det2Ver to perform robustly in complex zero-shot scenarios. The integration of these techniques allows the model to handle ZSL tasks effectively by leveraging the knowledge gained from related tasks.

B. Method Limitations

1) Cross-Domain Transferability Matter: While Det2Ver effectively leverages knowledge from the rumor detection task to enhance performance in fact verification, it fundamentally relies on data from different sources. The homogeneity or heterogeneity of the data significantly impacts the model's performance, especially concerning domain inconsistency, such as in the fields of biological science and news/social domains.

The ablation studies illustrated in *Domain Divergence of Rumor Detection DataSets*, Section V-C also indicate that

the data between the two tasks does not lead to performance improvement under inconsistent semantic distributions across different domains; in fact, it may even degrade fact verification performance. In response to this, in the future, we intend to rely more on rumor detection data that is both effective and covers a broader semantic context, thus mitigating the performance fluctuations caused by disparate data sources.

2) Potential Bias Analysis: The utilization of external knowledge from rumor detection datasets introduces several potential biases that need to be addressed. **Data Source Bias**: Different rumor detection datasets may come primarily from specific social media platforms, where the way information is spread and the content itself might exhibit systemic biases. Such biases could impair the model's performance when processing information from other platforms or sources. For instance, a dataset mainly sourced from Twitter might not enable the model to perform well on information from Facebook or news websites. **Language and Cultural Bias**: Linguistic and cultural differences in the datasets may also introduce biases. These biases can cause the model to exhibit skewed performance across different languages or cultures. For example, a model trained exclusively on English datasets might struggle to handle information effectively in other languages. **Sample Distribution Bias**: Imbalanced sample distributions within the datasets can further lead to biases. If a dataset contains a disproportionate number of true news samples compared to false news, the model may be biased towards predicting news as true, thus reducing its effectiveness in detecting false news. Addressing these biases requires careful consideration and targeted strategies to ensure the model's robustness and fairness. In future work, we plan to implement targeted strategies such as incorporating diverse and applying advanced bias detection to address these biases and further ensure fairness and accuracy.

3) Lack of Inherent Fact-Checking Mechanisms in GLLM: The GLLM used in Det2Ver, like many generative language models, does not have built-in mechanisms to verify the facts it generates. This can lead to the generation of inaccurate or misleading information, especially when the input data is ambiguous or contains misinformation.

To mitigate this, we plan to incorporate additional verification steps post-generation in future work. For instance, integrating fact-checking APIs or modules that cross-reference generated information with trusted databases can help validate the outputs. Additionally, human-in-the-loop approaches, where generated content is reviewed by experts or crowdworkers, can further ensure accuracy.

4) Amplification of Biases: GLLMs are trained on large datasets that may contain various biases, including cultural, social, and linguistic biases. These biases can be unintentionally amplified by the model, potentially perpetuating stereotypes or unfair representations.

To address this, we plan to employ several strategies in future work: 1. *Bias Detection and Mitigation*: Implement bias detection algorithms that identify and flag biased content during training and generation. Techniques such as adversarial training and bias regularization can be used to reduce biases in the model's outputs. 2. *Diverse Training Data*: Ensure the training data is as diverse and representative as possible, covering a wide range of perspectives and contexts to minimize inherent biases. 3. *Ethical Guidelines and Review*: Establish ethical guidelines for content generation and review the outputs regularly to identify and rectify any biased or unfair representations.

Furthermore, the quality of the training data significantly impacts the performance and fairness of the GLLM. Poorly curated or biased datasets can lead to suboptimal model performance and biased outputs. To address this issue, we plan to optimize the data sources in our future work, which includes the following steps: data cleaning, removing or correcting misleading, false, or biased data from the training corpus; and employing high-quality annotation standards to ensure consistency across the dataset. Through these measures, we aim to improve the quality of the training data, thereby enhancing the overall performance and fairness of our Det2Ver.

C. The Social Information Significance of Synergizing Rumor Detection and Fact Verification

In today's digital age, the rapid spread of misinformation on social media and other online platforms poses significant challenges. Misinformation can have severe consequences, ranging from public health risks to political instability. Traditional methods of rumor detection, which typically involve binary classification of information as true or false, often fall short in addressing the complexities of modern misinformation. Fact verification, on the other hand, involves a more detailed analysis, considering the context and evidence surrounding claims. By integrating rumor detection and fact verification, we can develop more comprehensive systems that not only identify misinformation but also provide contextual verification, enhancing the reliability of information in the digital landscape.

D. The Effects on Future Rumor Detection Research Area, and Real-World Applications

The method proposed in this work has significant implications for the fields of rumor detection and fact verification, as well as for broader societal applications. Firstly, our proposed approach significantly improves the accuracy and efficiency of rumor detection. Compared to existing methods, such as those relying on keyword matching or simple statistical models, which are prone to false positives or false negatives, our approach utilizes deep learning models and knowledge or supervision signal transfer between different tasks. This allows for more precise detection of subtle textual differences, thereby reducing false positive rates.

On social media platforms, our method can be used for real-time monitoring and detection of rumors, enabling platforms to promptly take measures to prevent the spread of misinformation. This not only helps protect users from being misled by misinformation but also maintains the platform's credibility. Additionally, government agencies and public health organizations can use our method to screen and verify public information, ensuring that the information disseminated to the public is accurate. For instance, during the pandemic,

our method can help quickly identify and stop the spread of misinformation about the outbreak, thereby safeguarding public health and social stability.

Furthermore, our research offers new directions and ideas for future studies. First, further research can explore how to extend our method to other languages or domains. Currently, our approach is primarily focused on English text, but future efforts could adapt it for rumor detection and fact verification in other languages. Additionally, future research could incorporate more diverse data sources, such as images, videos, and social network relationship data, to further enhance the accuracy and comprehensiveness of rumor detection and fact verification. For example, by integrating image recognition technology, it would be possible to detect and verify misinformation that includes images.

E. Ethical Considerations

In this work, we focus on developing low-resource fact verification system by providing user with efficient and accurate verification results, including evidence supporting or refuting fake news and misleading platform propaganda widely on the Internet. Our Det2Ver fact verification system is to assist users in automatically avoiding the misleading of misinformation and the risks associated with its dissemination. We see potential misuse of the fact verification system, especially in the fake news propaganda. For instance, there exists a potential risk that Det2Ver be abused to automatically detect information sources that support or contradict specific claims in narratives. This could lead to the embellishment or denigration of original statements.

Considering these considerations, it is crucial to responsibly develop the fact verification system to mitigate the risks of unintended consequences and uphold the integrity of information dissemination.

VII. Conclusion

Fact verification, evolved from fake new detection, has emerged as a emerging research area in language processing. Constructing supervised corpora for fact verification is labour-intensive, leading to a growing interest in low-resource learning approaches. The synergy between rumor detection and fact verification is not only innovative but also essential in today's rapidly evolving information environment. By leveraging the strengths of both tasks, we can create more robust and generalizable models capable of handling misinformation more effectively. This approach addresses the critical need for reliable information verification mechanisms in an age where the rapid spread of misinformation poses significant challenges to public trust and knowledge integrity.

Inspired by the similarities between fact verification and rumor detection, we investigate the cross-task synergies to enhance few-/zero-shot fact verification learning by utilizing supervisory signals from rumor detection. We introduce Det2Ver in this paper, a novel prompt-tuning framework that integrates fact verification and rumor detection, which employs several advanced strategies in prompting and answer engineering, enabling a unified approach to both tasks. Specifically, Det2Ver is a unified framework that enhances few-/zero-shot fact verification by leveraging cross-task knowledge transfer from rumor detection, employing innovative prompting and label synchronization mechanisms Experiments indicate that integrating rumor detection as a data augmentation significantly improves fact verification performance, especially in low-resource scenarios. We believe this work can serve as a pivotal milestone, paving a new pathway for future research in few-/zero-shot fact verification and potentially other linguistic tasks.

In our future work, we will explore more powerful LMs and their advanced semantic analysis capabilities, in order to more effectively utilize LMs as backbone networks for supporting cross-task knowledge transfer. Additionally, we plan to expand our scope of exploration to a wider range of domains, assessing the impact of different types of rumor detection data on the performance of our Det2Ver fact verification framework.

Acknowledgment

This work was conducted by W. Jin during his research at Xi'an Jiaotong University. Both authors, W. Jin and M. Jiang, contributed equally to this work. The corresponding author of this paper is G.Yang from Imperial College London. G. Yang was supported in part by the ERC IMI, UK (101005122), the H2020 (952172), the MRC, UK (MC/PC/21013), the Royal Society, UK (IEC\NSFC\211235), the NVIDIA Academic Hardware Grant Program, UK, the SABER project supported by Boehringer Ingelheim Ltd, UK, and the UKRI Future Leaders Fellowship, UK (MR/V023799/1). T. Tao was supported in part by the Key Program of the Natural Science Foundation of the Educational Department of Anhui Province of China (Grant No. 2022AH050319), and the University Synergy Innovation Program of Anhui Province (Grant No. GXXT-2023-021).

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APPENDIX A
CONSTRUCTED PROMPTING TEMPLATES

As illustrated in Table VII, we defined nine consolidation prompts, which are primary inspired by the XNLI task's prompts customized by PromptSource. XNLI (Cross-Lingual Natural Language Inference) [39] is an necessary benchmark task in Natural Language Understanding (NLU). Given two sentences, one as the premise and the other as the hypothesis, the XNLI task aims to determine the interrelationship (i.e. entailment, contradiction, and neutral) between this two sentences. PromptSource [40] is a comprehensive and high-quality system of natural language prompts. PromptSource encompasses over 2,000 specialized prompts ranging from roughly 170 datasets for users train and query LLMs.

Specifically, we transformed the prompts from PromptSource (originally used for three-category task, NLI) by revising the grammar and wording, thereby converting them into new prompts suitable for binary classification (Rumor Detection, `Yes/No`).

The third column shows the *Internal Prefix*, which is combined with the original hypothesis/claim, [HYPOTHESIS], to construct a new hypothesis/claim, [New_HYPO]. Following the intrinsic feature of fact verification, we constructed 'true', 'uncertain', and 'false', thereby obtaining three prompts with distinct semantics. This serves as a basis for further integration with Consolidation Prompts.

APPENDIX B
DET2VER DATA FEEDING

As outlined in Section III-A, to gain a clearer insight into how Det2Ver harmonizes the input formats of the two tasks via templating, we have delineated the specific methodologies with an illustrative example in Table VIII.

APPENDIX C
PRACTICAL IMPLEMENTATION EXAMPLE ANALYSIS

Let's consider the following practical implementation example for a rumor detection and fact verification scenario (refer to the Det2Ver framework in Figure 2 for further explanation):

- **Rumor (Hypothesis)**: "Eating carrots can significantly improve eyesight."
- **Premise**: "While vitamin A deficiency can lead to vision problems, most people obtain sufficient vitamin A through a balanced diet."

A. Conditional probability Loss Calculation

- **Input Sequence** X: "While vitamin A deficiency can lead to vision problems, most people obtain sufficient vitamin A through a balanced diet."
- **Target Sequence** Y: "Eating carrots can significantly improve eyesight."
- **Calculation**: For each token y_t in the target sequence, calculate $p(y_t \mid X, y_{<t})$, sum the log probabilities, and normalize by the sequence length T.

B. Unlikelihood Loss Calculation

- **Incorrect Target Sequence** $\hat{Y}$: "Eating carrots cannot improve eyesight."
- **Calculation**: For each token $y_t^{(i)}$ in the incorrect target sequence, calculate $1 - p(y_t^{(i)} \mid x, y_{<t}^{(i)})$, take the log, sum, and normalize by the number of incorrect target sequences.

C. Length Normalization Loss Calculation

- **Log Probability** $\delta(X, Y)$: Normalize the log probability of the correct target sequence "Eating carrots can significantly improve eyesight."
- **Incorrect Target Sequence Log Probability** $\delta(X, \hat{Y}^{(n)})$: Similarly normalize the log probabilities for all incorrect target sequences.

These calculations ensure that Det2Ver is trained not only to increase the probability of correct sequences but also to decrease the likelihood of incorrect ones, while accounting for the lengths of sequences to maintain robustness in predictions.

D. Label Words Synchronization Engine

To make the model's binary classification outputs suitable for fact verification, we apply the Label Words Synchronization Engine, which converts the intermediate results into the three classes: SUPPORT, REFUTE, and NOT ENOUGH INFORMATION (NEI).

For the given verification instance pair {premise, claim}, we first create three prompting variants corresponding to {true, uncertain, false}:

- **True (Support)**: "Based on the premise that most people obtain sufficient vitamin A through a balanced diet, is it true that eating carrots can significantly improve eyesight?"
- **Uncertain (NEI)**: "Based on the premise that most people obtain sufficient vitamin A through a balanced diet, is it uncertain that eating carrots can significantly improve eyesight?"
- **False (Refute)**: "Based on the premise that most people obtain sufficient vitamin A through a balanced diet, is it false that eating carrots can significantly improve eyesight?"

Based on these LM predictions, we obtain the language modeling probability distribution D_{LM} from the language model (LM) under each prompt.

1) Lookup Table-Based Retrieval: Subsequently, we use the Lookup Table-Based Retrieval method described in the model section to find the candidate answer $\hat{a}$ with the minimum logit: Logit_a for each prompt. This $\hat{a}$ is considered the final prediction result. Using the designed label mapping table, we map the intermediate results to the verification labels. For example, in this case, the prompts for Uncertain and True variants are mapped to "No", and the False variant is mapped to "Yes". Therefore, the hypothesis "Eating carrots can significantly improve eyesight" is refuted under the given premise.

TABLE VII
THE TEXTUAL DESCRIPTIONS OF OUR DESIGNED PROMPTING TEMPLATES DURING DET2VER'S PROMPTING CONSOLIDATION.

Serial Number	**Consolidation Prompts' Textual Descriptions**
T1	Taking the [PREMISE] into account, is it possible to ascertain the truth of the [New_HYPO]?
T2	[PREMISE] Using only the above PREMISE and what you know, can we induced that [New_HYPO] is true?
T3	Given that [PREMISE] \n Therefore, it must be true that [New_HYPO]? Yes or No?
T4	Suppose [PREMISE], Can we infer that [New_HYPO]? Yes or No?
T5	Considering the [PREMISE] and applying our knowledge, is it reasonable that [New_HYPO] is accurate?"
T6	[PREMISE] Keeping in mind the above text, consider: [New_HYPO] \n Is this true? Yes or No?
T7	In light of the [PREMISE], can we establish that the [New_HYPO] is indeed factual? Yes or No?
T8	[PREMISE] Based on that information, can we conject that the claim: [New_HYPO] is established?
T9	Based on the [PREMISE] provided and our understanding, can we infer that the [New_HYPO] holds true?
Internal Prefix	[**New_HYPO**] = "It is { `true` / `uncertain` / `false` } that" + [**HYPOTHESIS**]

TABLE VIII
THE DATA-FEEDING METHOD DESCRIPTION OF DET2VER DURING TRAINING/PREDICTION USING AN SPECIFIC EXAMPLE.

Rumor Detection	Det2Ver Input := `Rumor Detection Input`
	Rumor Detection Input: Eating carrots can significantly improve eyesight because, ... Det2Ver Input: Eating carrots can significantly improve eyesight because, ...
Fact Verification	Det2Ver Input := `Consolidation Prompting(Internal Prefix(Fact Verification Input))`
	Fact Verification Input: Claim: Eating carrots can significantly improve eyesight because, ... Premise: ... , most people can obtain sufficient vitamin A through a balanced diet.
	Det2Ver Inputs: 1. Suppose [..., most people can obtain sufficient vitamin A through a balanced diet.], Can we infer that It is true that [Eating carrots can significantly improve eyesight...]? Yes or No? 2. In light of the [..., most people can obtain sufficient vitamin A through a balanced diet.], can we establish that the It is uncertain that [Eating carrots can significantly improve eyesight...] is indeed factual? Yes or No? 3. Suppose [..., most people can obtain sufficient vitamin A through a balanced diet.], Can we infer that It is false that [Eating carrots can significantly improve eyesight...]? Yes or No?

2) Sequences Probability Ranking: However, during the lookup table matching process, there might be conflicting predictions that fall outside the table. For these unexpected predictions, we determine the verification label by assessing the fit ranking between the LM output probabilities and the decomposition sub-labels. For example, if all prompts (True, Uncertain, and False) for "Eating carrots can significantly improve eyesight" under the given premise result in No, we use this method to determine the final label.

By applying the Label Words Synchronization Engine, the model processes the example "Eating carrots cannot improve eyesight" and maps the intermediate outputs to one of the final verification labels: SUPPORT, REFUTE, or NEI, enhancing the overall performance and reliability of the verification process.